CLIMASCOPE AI

Climate Intelligence for El Niño and Indian Air Pollution

A Domain-Specific Language Model and Local AI Interface for Climate Science

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Base model: Qwen2.5-1.5B-Instruct
Fine-tuning method: Supervised Fine-Tuning
Parameter-efficient method: QLoRA
Original adapter: `airclimate_qlora_adapter`
Planned output directory: `ElNino_AirPollution_v4`
Local inference software: Ollama
User interface framework: Streamlit

Project Technical Report

Documentation of the current implementation, experimental configuration, and future development.

Abstract

CLIMASCOPE AI is a domain-focused artificial intelligence project designed to support questions and research related to El Niño, climate variability, atmospheric science, and air pollution in India.

The project combines a pretrained instruction-following language model with parameter-efficient fine-tuning and a lightweight conversational interface. The original fine-tuning workflow uses Qwen2.5-1.5B-Instruct and Quantized Low-Rank Adaptation (QLoRA), while the local computer has Qwen2.5 7B Instruct available through Ollama.

The original dataset contains question-and-answer examples organized into training, validation, and test splits. The initial split sizes are 4,475 training examples, 559 validation examples, and 560 test examples, giving a total of 5,594 examples.

A reduced experimental configuration was subsequently proposed to decrease training time by selecting 1,000 training examples, 200 validation examples, and 200 test examples. The completion and evaluation results of this reduced run must be documented after training finishes.

The interface, developed using Streamlit, provides a dark, futuristic environment for climate-related question answering. An Ollama API connection has been prepared for the locally available 7B base model. The separately trained 1.5B adapter must be integrated with its compatible base model before the interface can be considered a fine-tuned climate assistant.

This report describes the project's motivation, dataset, model configuration, fine-tuning approach, local deployment, interface, limitations, and proposed evaluation strategy.

Acknowledgements

This report documents the development of CLIMASCOPE AI, including the preparation of a question-and-answer dataset, the use of parameter-efficient fine-tuning techniques, and the development of a local conversational interface.

Contents

Abstract i

Acknowledgements ii

1 Introduction 1
- 1.1 Background . . . 1
- 1.2 Problem Statement . . . 1
- 1.3 Project Objectives . . . 1
- 1.4 Project Scope . . . 2

2 Scientific Background 3
- 2.1 El Niño and ENSO . . . 3
- 2.2 Indian Air Pollution . . . 3
- 2.3 Possible Links Between El Niño and Air Quality . . . 3
  - 2.3.1 Rainfall . . . 4
  - 2.3.2 Wind and Atmospheric Circulation . . . 4
  - 2.3.3 Temperature Inversions . . . 4
  - 2.3.4 Wildfires and Biomass Burning . . . 4
- 2.4 Scientific Qualification . . . 4

3 System Architecture 5
- 3.1 Overview . . . 5
- 3.2 Development Environment . . . 6

4 Dataset Preparation 7
- 4.1 Original Dataset . . . 7
- 4.2 Reduced Experimental Dataset . . . 7
- 4.3 Dataset Quality . . . 8
- 4.4 Instruction Formatting . . . 8

5 Model and Fine-Tuning Configuration 10
- 5.1 Base Model . . . 10

5.2 Supervised Fine-Tuning . . . 10
5.3 Quantized Low-Rank Adaptation . . . 10
5.4 Recorded Training Configuration . . . 11
5.5 Training Procedure . . . 12
5.6 Training Outputs . . . 12

6 Local Model Deployment 13
6.1 Local Environment . . . 13
6.2 Adapter Compatibility . . . 13
6.3 Ollama API . . . 14
6.4 Local Inference Status . . . 14

7 User Interface Design 15
7.1 Interface Overview . . . 15
7.2 Design Elements . . . 15
7.3 Research Domains . . . 15
7.4 Interface Implementation . . . 16
7.5 Running the Interface . . . 16

8 Evaluation Methodology 17
8.1 Purpose of Evaluation . . . 17
8.2 Proposed Evaluation Questions . . . 17
8.3 Evaluation Criteria . . . 18
8.4 Base Model Versus Fine-Tuned Model . . . 18
8.5 Quantitative Evaluation . . . 18

9 Current Progress and Findings 19
9.1 Completed or Confirmed Work . . . 19
9.2 Work Requiring Verification . . . 19
9.3 Technical Finding: Model Compatibility . . . 20
9.4 Scientific Finding: El Niño and Pollution . . . 20

10 Limitations 21
10.1 Dataset Limitations . . . 21
10.2 Model Limitations . . . 21
10.3 Causal Attribution Limitations . . . 21
10.4 Deployment Limitations . . . 21
10.5 Evaluation Limitations . . . 22

11 Future Work **23**
11.1 Complete and Verify Fine-Tuning 23
11.2 Improve Dataset Quality 23
11.3 Integrate the Adapter 23
11.4 Improve the User Interface 23
11.5 Add Evidence-Based Evaluation 24
11.6 Potential Research Extension 24

12 User Interface Showcase **25**

Conclusion **27**

List of Figures

3.1 Conceptual architecture of CLIMASCOPE AI. 5

12.1 CLIMASCOPE AI home screen and application layout. 25
12.2 Climate intelligence console and research-domain sidebar. 26
12.3 Example conversation about El Niño and Indian air pollution. 26
12.4 Local model response and the completed interface showcase. 26

1 Introduction

1.1 Background

Climate variability affects weather patterns, rainfall, temperature, and atmospheric circulation across different parts of the world. El Niño is one of the major components of the El Niño–Southern Oscillation (ENSO), a climate phenomenon associated with changes in tropical Pacific Ocean temperatures and atmospheric circulation.

These changes can influence regional weather conditions. In India, variations in rainfall, temperature, wind speed, and atmospheric stability can affect air quality. However, the relationship between El Niño and air pollution is complex and depends on the location, season, emissions, and other atmospheric and oceanic conditions.

CLIMASCOPE AI explores how a language model can be adapted to answer questions about these relationships in a clear, scientifically responsible manner.

1.2 Problem Statement

General-purpose language models can answer a wide range of questions, but their responses may not consistently provide the regional context, scientific qualifications, or specialized explanations required for climate research.

The project aims to develop a climate-focused assistant that can explain the relationship between El Niño and Indian air pollution, discuss atmospheric mechanisms, and help users formulate research questions.

1.3 Project Objectives

The main objectives are:

1. Develop a domain-focused climate question-answering assistant.
2. Explore supervised fine-tuning of Qwen2.5-1.5B-Instruct.
3. Use QLoRA to reduce the computational requirements of fine-tuning.

4. Prepare training, validation, and test datasets.
5. Develop a conversational interface using Streamlit.
6. Explore local inference through Ollama.
7. Evaluate answer quality, factual consistency, scientific uncertainty, and relevance to Indian air pollution.

1.4 Project Scope

The project focuses on four major subject areas:

- El Niño and ENSO.
- Indian air pollution and particulate matter.
- Climate variability and monsoon behaviour.
- Atmospheric science and environmental analysis.

The intended system is a research-support assistant. It is not a replacement for scientific measurements, official air-quality monitoring, or expert climate analysis.

2 Scientific Background

2.1 El Niño and ENSO

El Niño refers to the periodic warming of sea-surface temperatures in the central and eastern equatorial Pacific. It forms part of ENSO, which also includes La Niña and neutral conditions.

ENSO can influence atmospheric circulation and alter weather patterns in regions far from the tropical Pacific. The effects differ between events and are influenced by the strength and timing of the event and by interactions with other climate systems.

2.2 Indian Air Pollution

Air pollution in India results from multiple sources, including:

- Road transport and vehicle exhaust.
- Industrial activity and power generation.
- Construction activity and road dust.
- Biomass burning and residential fuel use.
- Agricultural residue burning in some regions and seasons.
- Secondary particles formed through atmospheric chemical reactions.

Fine particulate matter, commonly measured as PM2.5, consists of airborne particles with aerodynamic diameters of approximately 2.5 micrometres or less.

PM2.5 concentrations depend on emissions, atmospheric chemistry, mixing, wind, precipitation, and other meteorological conditions.

2.3 Possible Links Between El Niño and Air Quality

El Niño does not directly create all pollution. Instead, it may influence some of the meteorological conditions that affect pollution levels.

2.3.1 Rainfall

Rainfall can remove particles from the atmosphere through wet deposition. If rainfall is reduced during a particular period, this removal process may be weaker.

However, El Niño does not reduce rainfall everywhere or in every season. Regional effects must be investigated using observations.

2.3.2 Wind and Atmospheric Circulation

Wind transports and disperses pollutants. Changes in wind speed and direction may affect how pollutants accumulate or move between regions.

Weak ventilation can contribute to elevated pollution concentrations when emissions continue.

2.3.3 Temperature Inversions

During a temperature inversion, air near the ground can be cooler than the air above it. This structure can suppress vertical mixing and contribute to the accumulation of pollutants near the surface.

Inversions are influenced by local weather and seasonal conditions; they should not automatically be attributed to El Niño.

2.3.4 Wildfires and Biomass Burning

Hotter or drier conditions can increase fire risk in some regions. Smoke from fires can contribute to particulate pollution.

The magnitude and direction of this effect depend on regional climate, land management, fire activity, and emissions.

2.4 Scientific Qualification

A scientifically reliable assistant must distinguish between an established physical mechanism and a demonstrated regional relationship.

For example, atmospheric stagnation can contribute to pollutant accumulation. However, establishing that a specific El Niño event caused elevated PM2.5 in Delhi requires observational evidence and consideration of other possible explanations.

CLIMASCOPE AI is intended to communicate these distinctions rather than treating every El Niño event as a cause of severe air pollution.

3 System Architecture

3.1 Overview

The project has three principal components:

1. A pretrained base language model.
2. A domain-adaptation pipeline using SFT and QLoRA.
3. A conversational user interface for local inference.

The intended workflow is illustrated below.

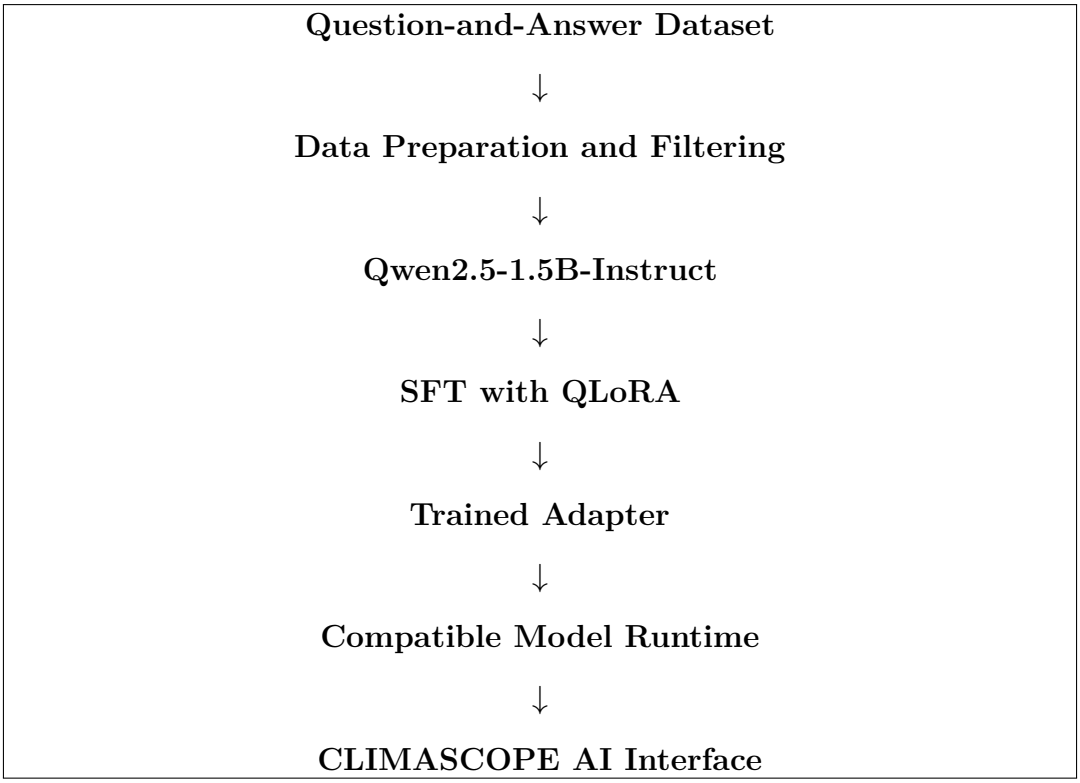


Figure 3.1: Conceptual architecture of CLIMASCOPE AI.

The diagram describes the intended complete workflow. The final adapter integration and inference path must be verified independently.

3.2 Development Environment

The project uses the following tools and frameworks.

Table 3.1: Software components

Component	**Purpose**
Google Colab	Model fine-tuning environment.
Python	Main programming language.
Hugging Face Transformers	Model loading, tokenization, and training.
PEFT	Parameter-efficient adapter training.
PyTorch	Model computation and training operations.
Datasets	Dataset loading and preprocessing.
Streamlit	Conversational user interface.
Ollama	Local serving of a compatible language model.
VS Code	Local development and code editing.

4 Dataset Preparation

4.1 Original Dataset

The original dataset is organized into three splits. Each example contains a question and its corresponding answer.

Table 4.1: Original dataset split sizes

Split	Number of examples
Training	4,475
Validation	559
Test	560
Total	5,594

The dataset columns reported during inspection were:

- `question`
- `answer`

The original splits were already separated into training, validation, and test data.

4.2 Reduced Experimental Dataset

To reduce the computational cost and duration of training, a smaller subset was proposed.

Table 4.2: Planned reduced dataset configuration

Split	Selected examples
Training	1,000
Validation	200
Test	200
Total selected	1,400

The selection procedure uses random shuffling with seed 42, followed by selection of the required number of examples from each split.

The reduced configuration represents a planned or attempted experiment. The actual dataset sizes used by the final completed training run should be verified from the Colab output before the results are finalized.

4.3 Dataset Quality

The dataset contains general climate-science questions alongside questions covering environmental and policy topics. For example, some examples address carbon taxation and radiative forcing.

A dataset containing broad climate information may help with general scientific language and explanations. However, it may not be sufficiently specialized for questions about El Niño and Indian air pollution.

The following improvements are recommended:

1. Identify questions that directly address ENSO, Indian air quality, PM2.5, monsoon rainfall, and atmospheric circulation.
2. Remove duplicate and near-duplicate examples.
3. Check answers for factual errors and unsupported causal claims.
4. Ensure that the validation and test sets are not used as training examples.
5. Preserve a separate test set for final evaluation.
6. Document the original source and licence of the dataset where available.

4.4 Instruction Formatting

Each training example is converted into a chat-style conversation with three messages:

1. A system message defining the assistant's role.
2. A user message containing the question.
3. An assistant message containing the answer.

The system prompt used in the preparation workflow is:

> You are a climate-science assistant specializing in El Niño, Indian air pollution, climate variability, and atmospheric science. Give clear, accurate, evidence-based answers. Do not invent facts. Acknowledge uncertainty when appropriate.

The conversation is formatted using the tokenizer's chat template where supported by the selected model.

5 Model and Fine-Tuning Configuration

5.1 Base Model

The base model selected for the fine-tuning experiment is Qwen2.5-1.5B-Instruct.

It is an instruction-following language model with approximately 1.5 billion parameters. The model provides the general language and reasoning capabilities on which the domain adaptation is built.

5.2 Supervised Fine-Tuning

Supervised Fine-Tuning (SFT) trains the model on examples containing an input instruction and a target response.

In this project, the user question serves as the input, while the corresponding climate-science answer serves as the target output.

The purpose is to adapt the model's response behaviour to the task and terminology of the target domain.

5.3 Quantized Low-Rank Adaptation

QLoRA combines a quantized base model with trainable low-rank adapter parameters.

Instead of updating every parameter in the pretrained model, the method introduces trainable low-rank matrices into selected model components.

For a weight update, the low-rank approximation can be expressed as:

$$W' = W + \Delta W$$

where

$$\Delta W = BA$$

and the matrices A and B have a lower effective rank than the original weight matrix.

The adapter is trained while the quantized base weights remain frozen. This can reduce the memory requirements compared with full-parameter fine-tuning.

5.4 Recorded Training Configuration

The following values were used in, or recorded for, the original training workflow.

Table 5.1: Fine-tuning configuration

Parameter	**Value**
Base model	Qwen2.5-1.5B-Instruct
Fine-tuning method	SFT with QLoRA
Quantization	4-bit
Quantization type	NF4
LoRA rank (`r`)	8
LoRA alpha	16
LoRA dropout	0.05
Learning rate	2×10^{-4}
Number of epochs	2
Training batch size per device	1
Evaluation batch size per device	1
Gradient accumulation steps	8
Effective batch size	8, assuming one device
Warmup ratio	0.03
Maximum gradient norm	0.3
Optimizer	`paged_adamw_8bit`
Gradient checkpointing	Enabled
Length grouping	Enabled in the recorded configuration
Logging interval	Every 5 steps in the original configuration
Random seed	42 in the reduced-data preparation cell
Original maximum sequence length	Intended to be 512; verify actual tokenization settings
Original adapter directory	`/content/airclimate_qlora_adapter`
New experiment output directory	`/content/ElNino_AirPollution_v4`

The effective batch size is the per-device batch size multiplied by the gradient accumu-

lation steps and the number of devices. For a single training device, the recorded settings give an effective batch size of eight.

Some values above describe the original training configuration, while others describe the reduced-data experiment. They should not be interpreted as proof that every listed setting was used in the final completed run.

5.5 Training Procedure

The intended training workflow is:

1. Load the base model and tokenizer.
2. Configure 4-bit quantization.
3. Prepare the LoRA configuration.
4. Convert questions and answers into chat-formatted training text.
5. Tokenize the training and validation examples.
6. Configure the trainer and data collator.
7. Train for the configured number of epochs.
8. Evaluate the trained model.
9. Save the adapter and tokenizer.

5.6 Training Outputs

The original adapter directory was named:

`airclimate_qlora_adapter`

The newer experiment was configured to use:

`/content/ElNino_AirPollution_v4`

A typical PEFT adapter directory contains:

- `adapter_config.json`
- `adapter_model.safetensors`, or an alternative supported weight file
- `tokenizer_config.json`, where saved
- Additional tokenizer files, depending on the save procedure

The exact contents must be checked in the downloaded archive. The adapter configuration identifies the base model and the adapter parameters.

6 Local Model Deployment

6.1 Local Environment

The local development environment uses Windows and Visual Studio Code. Ollama is installed and reports the following model entries:

Table 6.1: Models reported by the local Ollama installation

Model entry	Reported ID	Reported size
`qwen2.5:7b-instruct`	`845dbda0ea48`	4.7 GB
`qwen2.5:7b`	`845dbda0ea48`	4.7 GB

The two tags share the same reported model identifier. They therefore appear to refer to the same underlying model artifact rather than two separate model downloads.

The locally available model is Qwen2.5 7B Instruct. It is distinct from the Qwen2.5-1.5B-Instruct model used for the adapter-training workflow.

6.2 Adapter Compatibility

A LoRA adapter is associated with the base-model architecture and parameter structure for which it was trained.

The 1.5B adapter cannot simply be attached to the 7B model. Their parameter structures are not interchangeable.

There are two possible deployment directions:

1. Use the compatible Qwen2.5-1.5B base model with the trained adapter through a suitable Transformers and PEFT inference setup.

2. If a compatible conversion workflow is available, merge the adapter into the matching 1.5B base model and prepare the merged model for a supported local runtime.

The correct route depends on the available hardware, runtime support, and the actual files in the adapter archive.

6.3 Ollama API

Ollama exposes a local HTTP API. The interface can send a question to the locally available model through the chat endpoint.

The local endpoint used in the prototype is:

```
http://localhost:11434/api/chat
```

A request can contain a system instruction and the conversation history. The response contains the generated assistant message.

The API connection uses the 7B model by default in the current prototype. It does not automatically load the separately trained 1.5B adapter.

6.4 Local Inference Status

The following distinctions are important:

- Ollama is installed.
- The 7B model entries appear in the local model list.
- The Streamlit interface has been created.
- An Ollama API connection has been prepared in the interface code.
- Successful end-to-end inference through the UI should be verified by running the application and checking its response.
- Integration of the trained 1.5B adapter remains a separate task.

7 User Interface Design

7.1 Interface Overview

The application is named CLIMASCOPE AI. Its design uses a dark background, cyan highlights, and a simple conversation layout intended to resemble a climate intelligence console.

The initial prototype is implemented as a single Python file named `app.py` using Streamlit.

7.2 Design Elements

The prototype includes:

- Application title and climate-science subtitle.
- A sidebar showing the system status.
- A list of research domains.
- A conversational question input.
- A message history maintained using Streamlit session state.
- A placeholder response in the initial version.
- A planned Ollama API request for local generation.

7.3 Research Domains

The interface highlights four principal domains:

1. El Niño and ENSO.
2. Indian air pollution.
3. Climate variability.

4. Environmental analysis.

7.4 Interface Implementation

The interface uses the Streamlit configuration function to set the page title, icon, and layout.

Custom CSS is used to style the page background, sidebar, headings, buttons, and input field.

The conversation is maintained in `st.session_state`, allowing messages to remain visible during interaction with the application.

The chat input accepts a user question. In the initial version, the assistant returned a placeholder message explaining that the model had not yet been connected. A later version was prepared to send requests to the Ollama local API.

The application should be tested to confirm that the installed dependencies, API endpoint, and model tag match the actual local environment.

7.5 Running the Interface

The prototype is launched from the terminal with:

```
streamlit run app.py
```

The command starts the Streamlit development server and normally opens the interface in a web browser.

The interface is local, but the language model's behaviour depends on which model is actually configured and loaded by the inference runtime.

8 Evaluation Methodology

8.1 Purpose of Evaluation

A working user interface does not establish that a fine-tuned model is scientifically accurate. The model must be evaluated separately from the visual interface.

Evaluation should measure factual correctness, relevance, scientific reasoning, uncertainty handling, and the ability to distinguish correlation from causation.

8.2 Proposed Evaluation Questions

The following questions can be used for an initial qualitative assessment.

1. How does El Niño influence air pollution in India, particularly PM2.5 concentrations in Delhi during winter? Explain the atmospheric mechanisms, rainfall and wind patterns, and interactions with other climate factors. Distinguish established findings from uncertain or region-dependent effects.

2. How does El Niño affect India's monsoon rainfall, and why do the effects vary across different regions?

3. Why does Delhi experience severe PM2.5 pollution during winter, and how can temperature inversions restrict atmospheric mixing?

4. Can El Niño be used to predict India's winter air pollution levels? What observations would be required for a reliable prediction?

5. Design a research methodology to investigate whether El Niño events are associated with increased winter PM2.5 concentrations across northern India. Identify the datasets, confounding factors, statistical methods, and limitations.

6. Does every El Niño event cause severe air pollution in Delhi? Explain the evidence and limitations behind the answer.

8.3 Evaluation Criteria

Each response can be assessed against the following criteria.

Table 8.1: Proposed qualitative evaluation rubric

Criterion	**Assessment question**
Factual accuracy	Are the scientific claims correct?
Domain relevance	Does the answer address El Niño and Indian air pollution?
Mechanistic explanation	Does it explain relevant atmospheric processes?
Uncertainty	Does it acknowledge regional and seasonal variability?
Causal reasoning	Does it avoid treating association as proof of causation?
Completeness	Does it cover the important aspects of the question?
Clarity	Is the answer understandable and well organized?

8.4 Base Model Versus Fine-Tuned Model

A meaningful experiment should compare the base model and fine-tuned model using the same test questions.

The base model provides a reference point. The fine-tuned model can then be assessed to determine whether adaptation improves domain relevance, scientific consistency, or response quality.

No performance improvement should be claimed until the outputs have been evaluated.

8.5 Quantitative Evaluation

If a reliable set of reference answers is available, additional metrics may be used. However, text similarity alone is not sufficient for assessing scientific correctness.

A useful evaluation process combines automated metrics, where appropriate, with human review of factual accuracy, causal reasoning, and uncertainty.

For a research-oriented application, the evaluation questions and scoring criteria should be fixed before comparing the models to reduce bias.

9 Current Progress and Findings

9.1 Completed or Confirmed Work

Based on the recorded development steps, the following work has been completed or confirmed:

1. The original question-and-answer dataset was inspected, and its training, validation, and test split sizes were identified.
2. A climate-science system prompt was prepared.
3. A QLoRA fine-tuning configuration was defined for Qwen2.5-1.5B-Instruct.
4. An original adapter archive named `airclimate_qlora_adapter` is available.
5. The local Ollama installation reports the Qwen2.5 7B Instruct model.
6. A basic Streamlit interface named `app.py` was created.
7. A reduced-dataset configuration was prepared to shorten the training process.

9.2 Work Requiring Verification

The following items should not be treated as completed without additional evidence:

1. Whether the reduced 1,000-example training run completed successfully.
2. Whether the final adapter and tokenizer were saved correctly.
3. The final training loss and validation metrics.
4. Whether the Ollama-connected interface generates responses successfully on the local machine.
5. Whether the trained 1.5B adapter has been loaded into a compatible inference pipeline.
6. Whether fine-tuning improves scientific answer quality compared with the base model.

9.3 Technical Finding: Model Compatibility

An important implementation finding is that the locally available 7B model and the trained 1.5B adapter belong to different model sizes.

The adapter cannot be used with the 7B model simply by changing the model name in the interface. A compatible base model or an appropriate conversion and deployment procedure is required.

9.4 Scientific Finding: El Niño and Pollution

El Niño is not a direct or universal cause of air pollution. It can influence weather patterns that affect pollutant dispersion, rainfall, and fire risk in some regions.

The resulting effect on Indian air quality depends on regional meteorology, the timing and strength of ENSO, local and transported emissions, and other climate factors.

A causal claim about a particular Indian pollution episode requires observational analysis rather than relying on the existence of an El Niño event alone.

10 Limitations

10.1 Dataset Limitations

The original dataset contains general climate-science material and may not consistently focus on the relationship between El Niño and Indian air pollution.

A reduced sample can speed up experimentation, but it may also reduce coverage of less common topics. The effects depend on how representative the selected examples are.

10.2 Model Limitations

A language model can produce plausible but incorrect answers. Fine-tuning does not guarantee factual accuracy, and the model may reproduce errors present in the training data.

The model's output should therefore be treated as research assistance rather than authoritative scientific evidence.

10.3 Causal Attribution Limitations

Air pollution is affected by emissions, atmospheric chemistry, weather, regional transport, and seasonal behaviour.

Establishing the effect of El Niño requires separating its influence from other variables. A conversational model cannot establish this causal relationship merely by generating an explanation.

10.4 Deployment Limitations

The original adapter was trained for the 1.5B model, while the local Ollama installation reports the 7B model.

The adapter and runtime must be made compatible before the fine-tuned model can be used locally.

Performance also depends on the computer's hardware, available memory, and inference configuration.

10.5 Evaluation Limitations

No final numerical benchmark results are reported in this document because verified training metrics and controlled base-versus-fine-tuned evaluation results have not been provided.

This avoids presenting unverified performance figures as experimental findings.

11 Future Work

The following development steps are proposed.

11.1 Complete and Verify Fine-Tuning

Complete the reduced-data training run and record:

- Actual training and validation sample counts.
- Final training loss.
- Final validation loss or other evaluation metrics.
- Training duration.
- Final adapter directory and file names.

11.2 Improve Dataset Quality

Review the data for duplicate questions, unsupported answers, and insufficient domain coverage. Add reliable, properly sourced examples that directly address ENSO, Indian air pollution, PM2.5, rainfall, and atmospheric circulation.

11.3 Integrate the Adapter

Load the adapter using its compatible Qwen2.5-1.5B base model. Test generation independently before connecting the model to the interface.

11.4 Improve the User Interface

Potential improvements include:

- A climate-research conversation history.

- Clear model and adapter status indicators.
- A dedicated research-question mode.
- Structured explanations of mechanisms and uncertainty.
- Optional evidence and source panels.
- Better handling of long responses and errors.

11.5 Add Evidence-Based Evaluation

Compare the base model and fine-tuned model using a fixed test set. Document evaluation criteria and provide representative examples of successful and unsuccessful answers.

Where possible, use trustworthy scientific literature and observational datasets to verify the answers.

11.6 Potential Research Extension

A future version could investigate relationships between ENSO indices and observed PM2.5 concentrations across Indian cities.

Such an analysis could combine ENSO indices, air-quality observations, meteorological variables, and emission proxies. Statistical modelling would be required to account for seasonality, trends, and confounding factors.

The language model could assist with explaining the results, but the scientific conclusions would need to come from the data and analysis.

12 User Interface Showcase

This chapter provides space for screenshots of the CLIMASCOPE AI interface. The images can document the application's visual design, conversation interface, local model response, and future fine-tuned model integration.

Replace the placeholders with actual screenshots from the application. The captions can be edited to describe the specific content shown in each image.

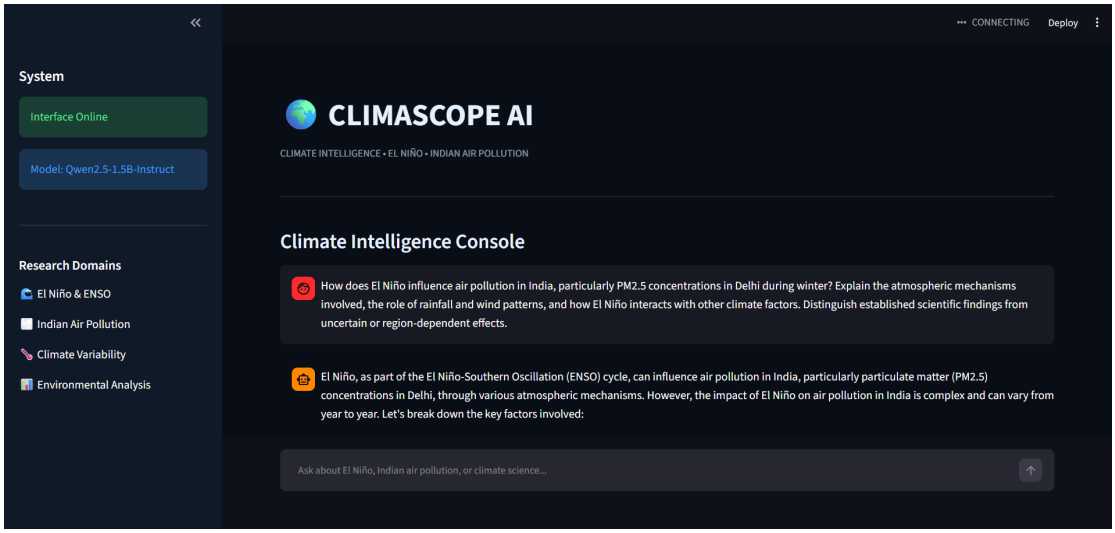


Figure 12.1: CLIMASCOPE AI home screen and application layout.

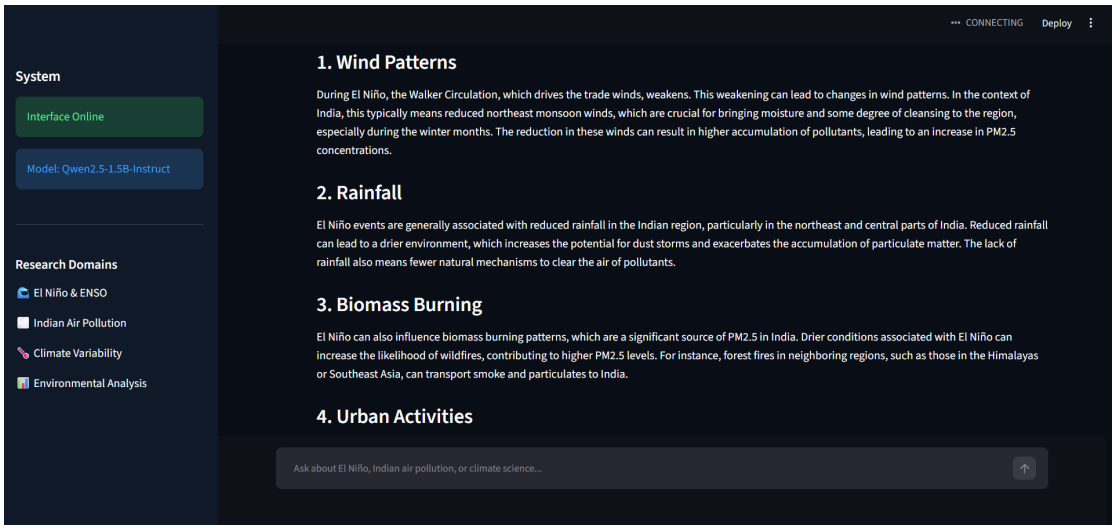


Figure 12.2: Climate intelligence console and research-domain sidebar.



Figure 12.3: Example conversation about El Niño and Indian air pollution.

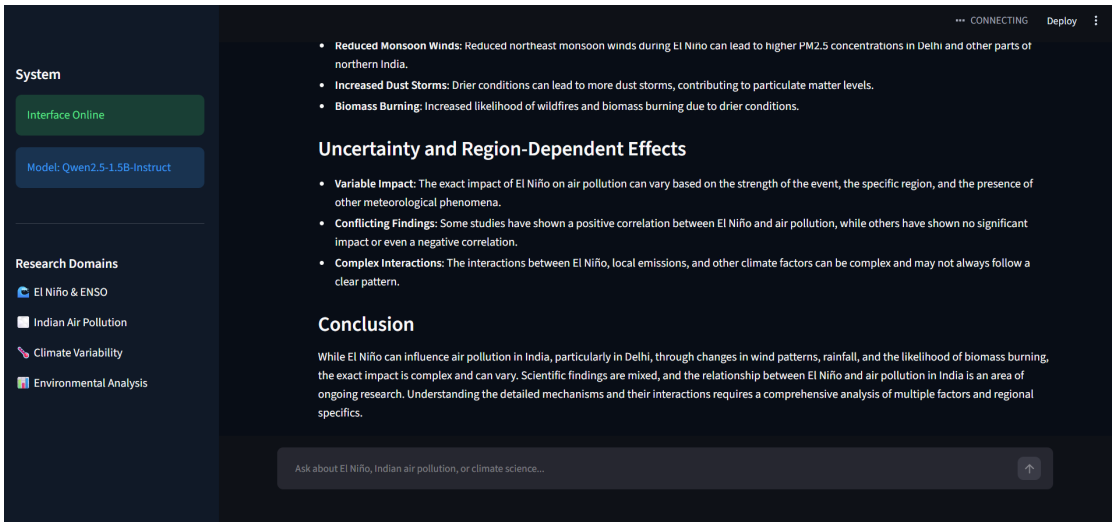


Figure 12.4: Local model response and the completed interface showcase.

Conclusion

CLIMASCOPE AI explores the development of a domain-focused language model assistant for questions concerning El Niño, climate variability, atmospheric science, and Indian air pollution.

The project brings together a question-and-answer dataset, a Qwen2.5-1.5B-Instruct fine-tuning workflow, QLoRA configuration, and a Streamlit conversational interface. The local environment also contains Qwen2.5 7B Instruct through Ollama, providing a route for local language-model inference.

An important technical distinction is that the locally available 7B model is not directly compatible with the adapter trained for the 1.5B model. Correct integration of the adapter with its matching base model remains necessary for the interface to use the fine-tuned assistant.

The scientific motivation is to help users understand how El Niño may influence weather patterns that affect air quality without incorrectly treating it as the sole cause of pollution. Reliable analysis requires regional observations, careful treatment of confounding factors, and appropriate uncertainty.

The next stages are to verify the final training run, test the adapter, integrate the compatible model into the interface, and evaluate its answers against a fixed set of scientific questions.

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